

DOCUMENT 2.2 — TECHNICAL ARCHITECTURE & ERD DIAGRAM

Folder 2: Project Planning Phase | ServiceNow Capstone Project Submission

Project Title: Hospital Appointment Management System

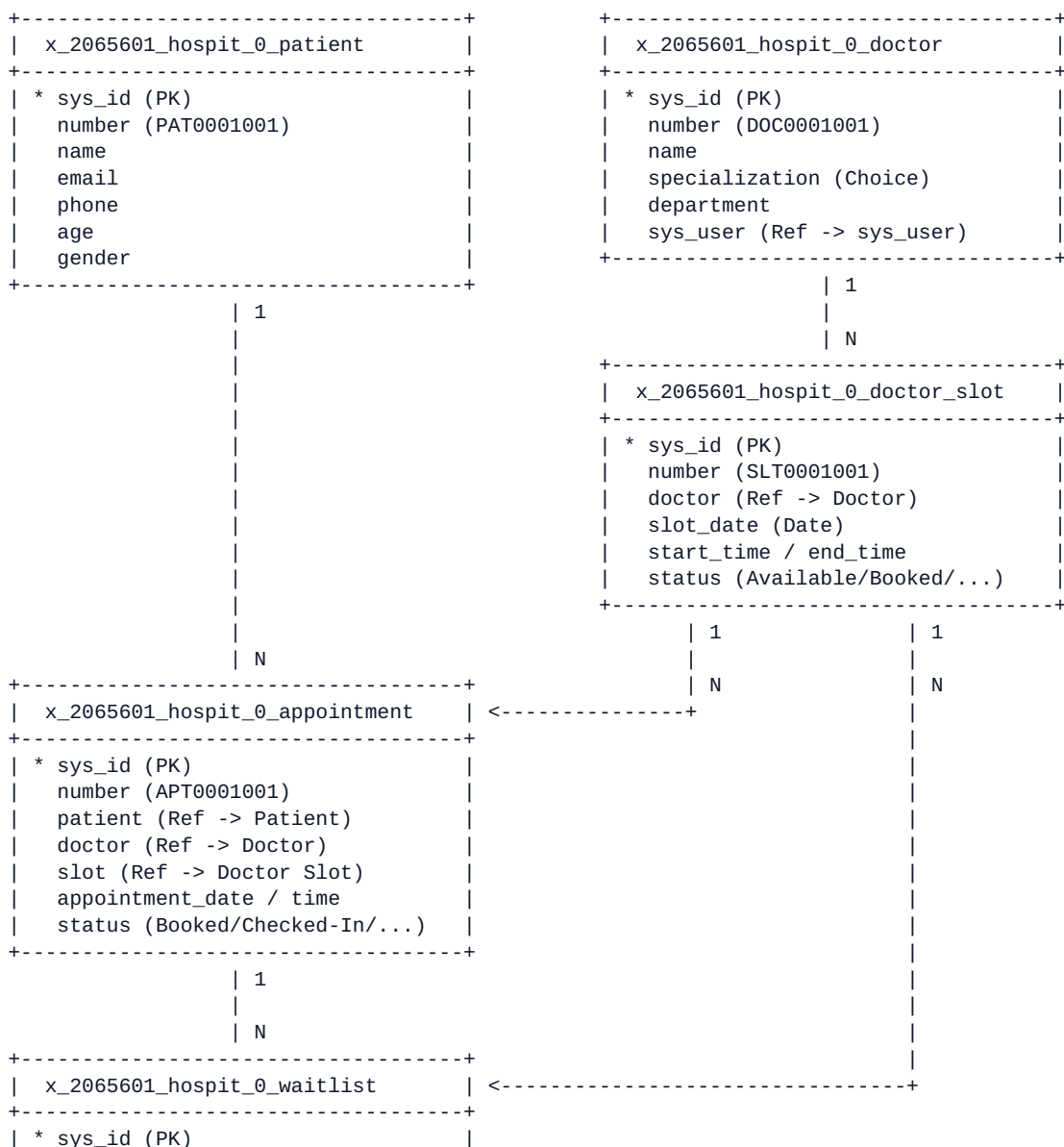
Platform: ServiceNow (Scoped Application)

Application Scope: x_2065601_hospit_0

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1. Entity Relationship Diagram (ERD)

The application architecture is structured around five custom tables in application scope `x_2065601_hospit_0`. Strict relational integrity and foreign key reference relationships link patients, doctors, doctor slots, appointments, and waitlists.



```

| number (WTL0001001) |
| patient (Ref -> Patient) |
| doctor (Ref -> Doctor) |
| slot (Ref -> Doctor Slot) |
| priority_order (Integer) |
| status (Waiting/Promoted/...) |
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```

2. Detailed Data Dictionary & Schema Definitions

Table Name	Field Name	Type	Key Reference / Choices & Details
x_2065601_hospit_0_patient	number	String	Auto-numbered PK (e.g., PAT0001001)
	name , email , phone	String	Patient demographic contact information
	age , gender	Integer / Choice	Clinical profile details
x_2065601_hospit_0_doctor	number	String	Auto-numbered PK (e.g., DOC0001001)
	specialization	Choice	Cardiology, Neurology, Pediatrics, Orthopedics, General
	sys_user	Reference	References platform sys_user table for authentication
x_2065601_hospit_0_doctor_slot	number	String	Auto-numbered PK (e.g., SLT0001001)
	doctor	Reference	References x_2065601_hospit_0_doctor
	status	Choice	Available , Booked , Completed , Cancelled
x_2065601_hospit_0_appointment	number	String	Auto-numbered PK (e.g., APT0001001)
	patient , doctor , slot	Reference	Links Patient, Doctor, and specific Doctor Slot
	status	Choice	Booked , Checked-In , Completed , Cancelled , No-Show
x_2065601_hospit_0_waitlist	number	String	Auto-numbered PK (e.g., WTL0001001)
	patient , doctor , slot	Reference	Links waiting patient to target doctor and full slot
	priority_order , status	Integer / Choice	Priority rank; Choices: Waiting , Promoted , Cancelled

3. System Architecture & Component Interaction

- **UI Layer (Service Portal):** Single-page AngularJS application widget (hospital_booking) handling multi-tab interactions (Home, Registration, Slot Booking, Doctor Queue, Waitlist, Admin).
- **Server Integration Layer (Script Include):** Core processing logic implemented in Script Include HospitalBookingUtils to execute atomic transactions (slot availability check + appointment creation).

- **Event-Driven Workflow Engine (Business Rules):** Business Rule `Auto Promote waitlist` triggers on `after update` when an appointment status changes to `Cancelled`, automatically promoting the highest priority waitlisted patient.